

SEMESTER END EXAMINATIONS - AUGUST / SEPTEMBER 2023

B.E. - Artificial Intelligence and					
Program	:	Machine Learning / Artificial Intelligence and Data Science	Semester	:	IV
Course Name	:	Theory of Computation	Max. Marks	:	100
Course Code	:	AI44 / AD44	Duration	:	3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define the following terms with a suitable example. i) Power of an Alphabet, ii) Length of a string, iii) Empty String, iv) Language, v) Empty Language. CO1 (05)
 - Design a DFA to accept each of the following languages: CO2 (05)
 - $L = \{w \in \{a,b\}^*: w \text{ has all strings that ends with substring } abb\}$.
 - $L = \{w; \text{ where } |w| \bmod 3 == 0, \text{ where } \Sigma = \{a\}\}$.
 - Convert to a DFA the following NFA in Fig. 1(c) and informally describe the language the automaton accepts. CO2 (10)

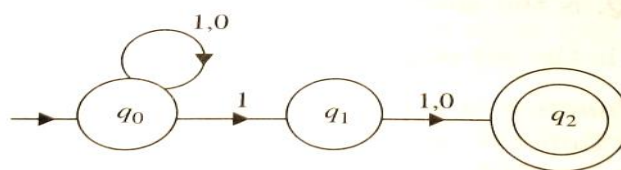


Fig. 1(c): NFA

- Differentiate between DFA, NFA and epsilon – NFA with respect to their acceptance of Language. CO1 (05)
 - Design a DFA to accept the following language: CO2 (05)
 - $L = \{w \mid w \text{ is of even length and begins with } 01 \text{ over } \Sigma = \{0,1\}\}$
 - $L = \{w \mid \text{the set of all strings that begin with } 01 \text{ and end with } 11. \text{ Over } \Sigma = \{0,1\}\}$.
 - Consider the following ϵ - NFA: CO2 (10)

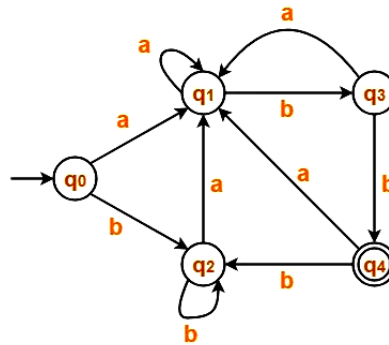
δ	ϵ	a	b
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p, r\}$
q	$\emptyset$	$\{p\}$	$\emptyset$
$*r$	$\{p, q\}$	$\{r\}$	$\{p\}$

- Compute the ϵ – closure of each state.
- Give the set of all strings of length 3 or less accepted by the automaton.
- Convert the automaton to DFA.

UNIT - II

- Define Regular Expression. Write Regular Expression for the following languages: CO2 (08)
 - $L = \{a^n b^m \mid m + n \text{ is even}\}$
 - $L = \{0^n 1^m \mid m \geq 1, n \geq 1, mn \geq 3\}$
 - State and prove pumping lemma theorem for regular language. CO2 (08)
 - Prove that regular languages are closed under complement/operation. CO2 (04)

4. a) Define distinguishable and indistinguishable states. Minimize the CO2 (10)
following Deterministic Finite State Machine (DFSM).



- b) Use the FA to regular expression conversion algorithm to construct a CO2 (10)
regular expression that describes $L(m)$.

δ	a	b
$\rightarrow^* 1$	2	$\{\}$
$*2$	3	1
3	3	1

UNIT - III

5. a) Define context-free Grammar for CO2 (06)
i) $L = \{0^i 1^j 2^k \mid j > i + k\}$
ii) $L = \{w \mid w \in \{a, b\}^*, na(w) \neq nb(w)\}$
b) Let $G = (V, T, P, S)$ be a CFG. Prove that if the recursive inference CO4 (07)
procedure tell us that terminal string w is in the language of variable A ,
then there is a parse tree with root A and yield w .
c) Construct an equivalent PDA for the grammar: CO2 (07)
 $S \rightarrow aABC$
 $A \rightarrow aB \mid a$
 $B \rightarrow bA \mid b$
 $C \rightarrow a$
6. a) Find leftmost and rightmost derivations, and a parse tree for the string CO2 (07)
 $+*-xyxy$ using the grammar:
 $E \rightarrow +EE \mid *EE \mid -EE \mid x \mid y$
b) Construct PDA to accept strings of the language $L = \{a^n b^n \mid n \geq 1\}$ and CO2 (08)
show moves for the string $aaabbbb$.
c) Define the following:- CO4 (05)
i) Language of CFG ii) Sentential form.

UNIT- IV

7. a) Define CNF. Convert the following CFG to CNF: CO4 (10)
 $S \rightarrow aACa$
 $A \rightarrow B/a$
 $B \rightarrow C/c$
 $C \rightarrow cC/c$
b) Design a PDA to accept the following language: CO4 (10)
 $L = \{a^n b^n; n \geq 0\}$
Draw the transition diagram for the constructed PDA. Show the IDs for
the string 'aaabbb'.
8. a) Differentiate between CNF and GNF with appropriate example. CO4 (05)
b) Design a PDA for the language that accepts the string with $n_a(w) < n_b(w)$ CO4 (10)
where $w \in (a + b)^*$ and show the instantaneous description of the PDA on
input 'abbab'
c) State and prove the pumping lemma for Context-Free languages. CO4 (05)

UNIT - V

9. a) Construct Turing machine for the language $L = \{0^n 1^n 2^n\}$. Show the instantaneous description for the string 000111222. CO3 (08)
b) Prove that if L is a recursive language, so is $\bar{L}$. CO4 (06)
c) Define post correspondence problem and also give suitable example. CO3 (06)
10. a) Define recursive languages. Prove that L_{ne} is recursively enumerable. CO4 (06)
b) Define a Turing machine and its instantaneous description. CO1 (04)
c) Design a Turing machine to implement the function for multiplication. CO3 (10)
Show the moves for 0010001.
